D Liu

About Me

Professional Interests: High Performance Computing, Data Engineering, Distributed Systems, Computational Nuclear Physics
Languages: Python, C++, C (GCC/clang), Java, Matlab, Fortran (f95), JavaScript
Technologies: Git, Unix, Shell, AWS, Jupyter/pandas, matplotlib, XGBoost, LightGBM, PostgreSQL, MCNP & Serpent
Legal Status: Canadian Citizen. Available on TN visa without sponsorship, or H1B.

Experience

Terrestrial Energy
Toronto
Computational Physics – Software Engineer
January 2020 – Present

- Developed core physics parameters for a next-generation nuclear reactor using monte carlo codes in Python & C
- Updated legacy simulation programs to support CPU/GPU multi-threading by adding code directives (OpenMP/OpenACC)
- Extended high performance concurrent C++ utilities to couple rapidly updating physical data sets for nuclear simulation
- Developed a full-scope simulator by integrating mathematical first principles on nuclear phenomena using Python

Ontario Power Generation
Toronto
Software Engineering, Reactor Simulations
September 2018 – December 2019

- Developed a heavy water decontamination scheduler for all nuclear reactors in Canada using VBA & Python
- Created a real time Python service that tracks and publishes logic decisions for plant control
- Improved large dataset performance and core features for preventative maintainence using WebSockets, C++ and JavaScript
- Optimized SQL database queries on multiple ODBC connections to push maintenance notices on critical nuclear equipment

Canadian Nuclear Laboratories
Chalk River
Software Engineer, Nuclear Simulations
January 2018 – September 2018

- Designed code to model the flow of groundwater surrounding below-grade nuclear waste repositories
- Implemented additional modules describing the interaction of dissolved chemical species with the surrounding earth
- Optimized aggregation and filter queries performed on virtual tables by strategically rejecting rows before computing expensive requests, using generators to reduce High Performance Compute memory usage by 80%
- Created MATLAB code to simulate gas-catalyst interactions to develop and scale-up an experimental process to separate Tritium from other gasses, a radioactive isotope of Hydrogen used for fusion energy research.

Projects and Activities

Chrome Extension ML Translator (WIP): Using Nvidia's Jarvis Automatic Speech Recognition package to convert foreign speech in youtube videos to subtitles for new language learners in realtime

10-Q/10-K Insights : Created a scraper in Python to acquire market quarterly and annual earning reports to predict future returns by detecting forward-looking risk statements from the Securities Exchange Commission

McKinsey & Company Hackathon : Analyzed if a requested loan is approved or not, based on the features of the applicant (e.g., income, gender, city), and that of the loan (e.g., loan amounts, interest rates). Used XGBoost / gradient boosting for analysis

Scotiabank Hackathon : Concepted an addicting mobile game to help reduce user debt with in-app purchases routed back to saving accounts. Participated with seven software engineers and graphics designers within 48 hours to win a $10,000 prize

Education

University of Ontario Institute of Technology *2018*
Bachelor of Engineering – Nuclear Engineering, Minors in Computer Science (95% cGPA)

Research : Created a MATLAB application to track AR device user location in GPS denied environments using MEMS input sensor data.